



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Date: 09/07/2026

Esmilna – Python Notebooks & Files

<https://github.com/Esmilna/-cuadernos-y-archivos-Python>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**

- ◆ REST API & Web Scraping
- ◆ Data Wrangling
- ◆ EDA & Visualization
- ◆ SQL Analysis
- ◆ Folium
- ◆ Plotly Dash
- ◆ Classification Models

- **Summary of all results**

- ◆ Launch-site performance
- ◆ Payload & orbit analysis
- ◆ Landing outcomes
- ◆ Geographic insights
- ◆ Interactive dashboard findings
- ◆ Classification performance
- ◆ Best-performing model

Introduction

- **Project background and context**

-  SpaceX Falcon 9

-  Historical launch data

-  Data Wrangling

-  EDA & SQL

-  Folium & Dash

-  Classification

- **Problems you want to find answers**

- Launch sites & success

- Payload & orbit

- Success over time

- Booster performance

- Landing outcomes

- Geographic relationships

- Best classification model

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

SpaceX API



Retrieve Launch Data



JSON Response



Parse JSON with Python



Create Pandas DataFrame



Collected Dataset

- **Source:** SpaceX REST API
- **Format:** JSON
- **Request:** HTTP GET
- **Tool:** Python `requests`
- **Processing:** JSON parsing
- **Storage:** Pandas DataFrame

- Data obtained from the **SpaceX REST API**
- HTTP requests used to retrieve launch records
- API response received in **JSON format**
- JSON data parsed using **Python**
- Data organized into a **Pandas DataFrame**
- Dataset prepared for subsequent **data wrangling and analysis**

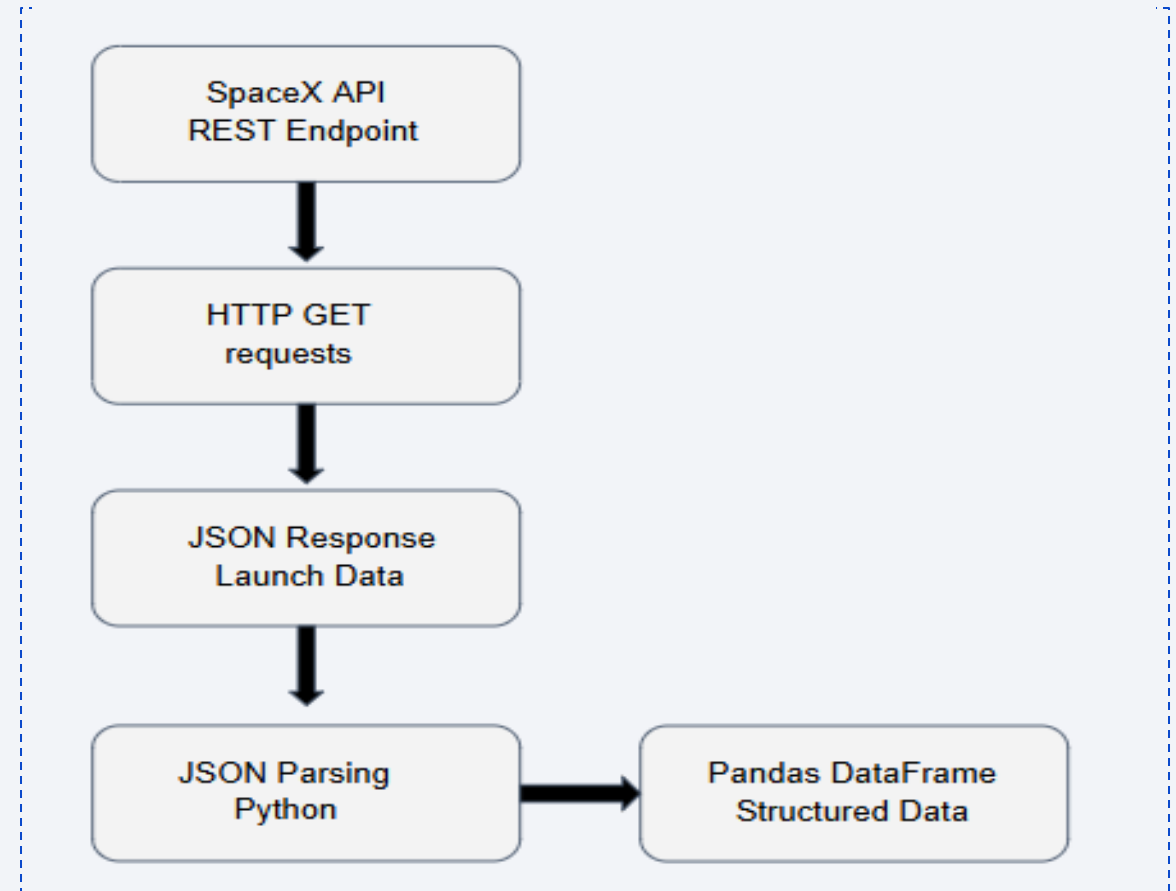
Data Collection – SpaceX API

- **Source:** SpaceX REST API
- **Format:** JSON
- **Request:** HTTP GET
- **Tool:** Python `requests`
- **Processing:** JSON parsing
- **Storage:** Pandas DataFrame

SpaceX Data Collection API

Jupyter Lab: *SpaceX Data Collection using API*

[View Notebook on GitHub](#)



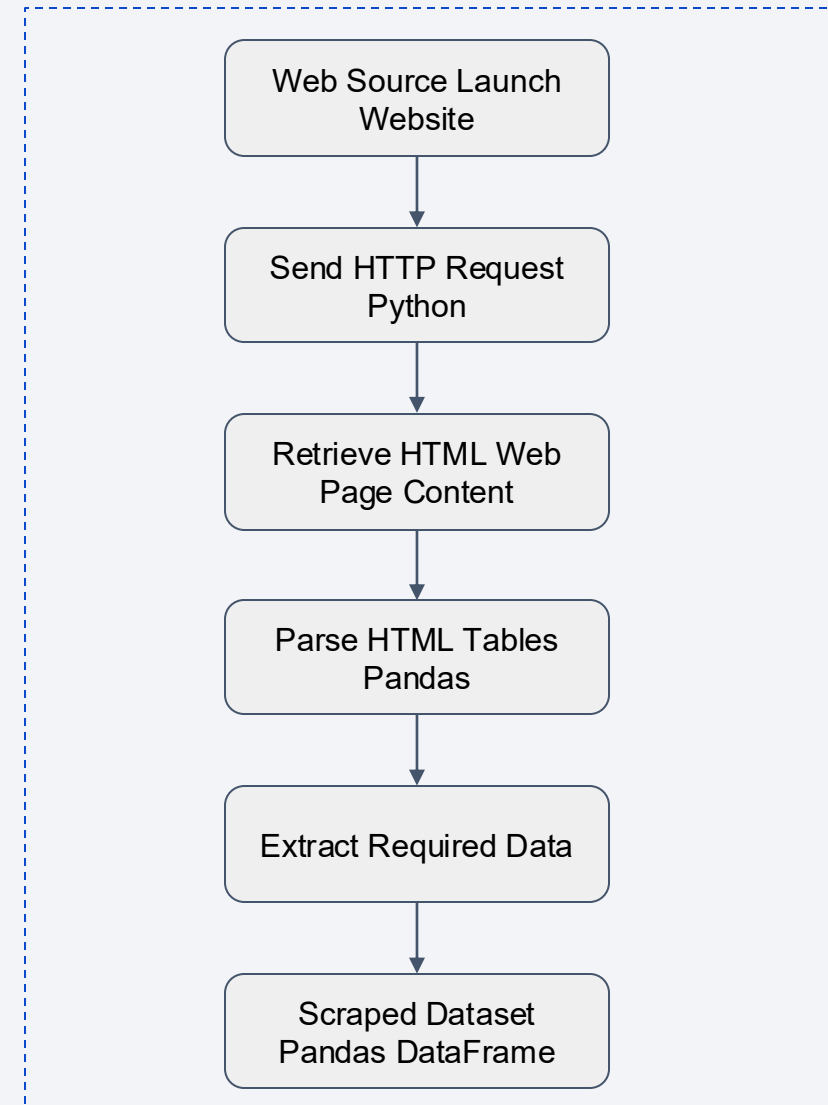
Data Collection - Scraping

- **Source:** Launch website
- **Method:** Web scraping
- **Request:** HTTP request
- **Content:** HTML
- **Parsing:** HTML tables
- **Tool:** Python / Pandas
- **Extraction:** Relevant launch records
- **Output:** Structured DataFrame

Web Scraping

Jupyter Lab: *SpaceX Web Scraping*

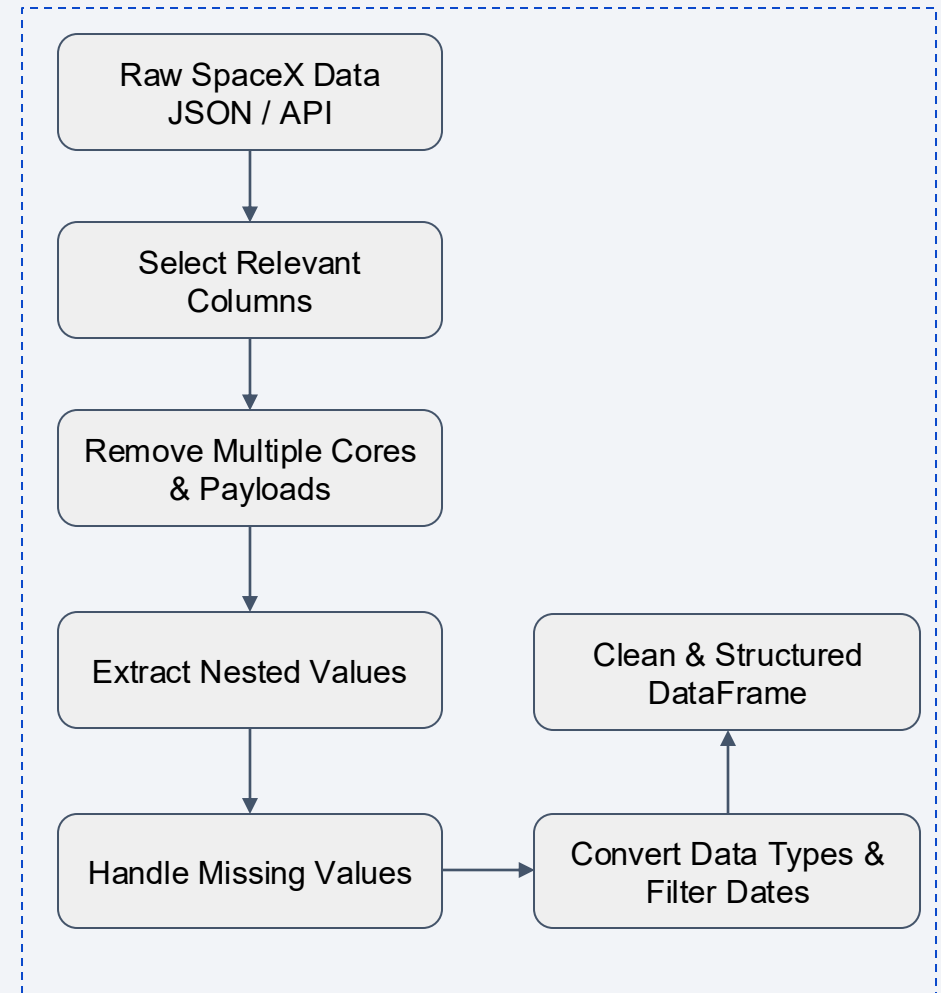
[View Notebook on GitHub](#)



Data Wrangling

Data Processing

- Column Selection
- Filter Multiple Cores/Payloads
- Extract Core & Payload IDs
- Extract Booster Version
- Convert `date_utc` to Datetime
- Filter Launch Date
- Calculate/Replace Missing `PayloadMass`
- Final Clean DataFrame



EDA with Data Visualization

- Analyze launch trends and mission outcomes
- Explore relationships between numerical variables
- Compare launch performance across launch sites
- Examine payload-related patterns
- Identify distributions and potential outliers
- Use visualizations to support data-driven insights

EDA with SQL

SQL Queries

- GROUP BY & COUNT()
- Subqueries & MAX()
- Date/string manipulation
- RANK() window function
- Landing outcome analysis

Advanced SQL Analysis

- GROUP BY & COUNT()
- Subqueries & MAX()
- Date/string manipulation
- RANK() window function
- Landing outcome analysis

Build an Interactive Map with Folium

Map Objects	Purpose	Result
Markers / Circles	Locate launch sites	Geographic visualization
MarkersCluster	Group launches	Better readability
Popups	Show information	Interactive exploration
PolyLine	Show distances	Proximity analysis
MousePosition	Get coordinates	Geographic inspection
Divicon	Show labels/distances	Enhanced map interpretation

Build a Dashboard with Plotly Dash

Plots

- Pie Chart
- Scatter Plot

Interactions

- Launch Site Dropdown
- Payload Range Slider
- Dynamic Callbacks

Interactive Dashboard

SpaceX Launch Dataset



Plotly Dash Dashboard



Select Launch Site +
Select Payload Range



Dash Callback Functions



Update Visualizations

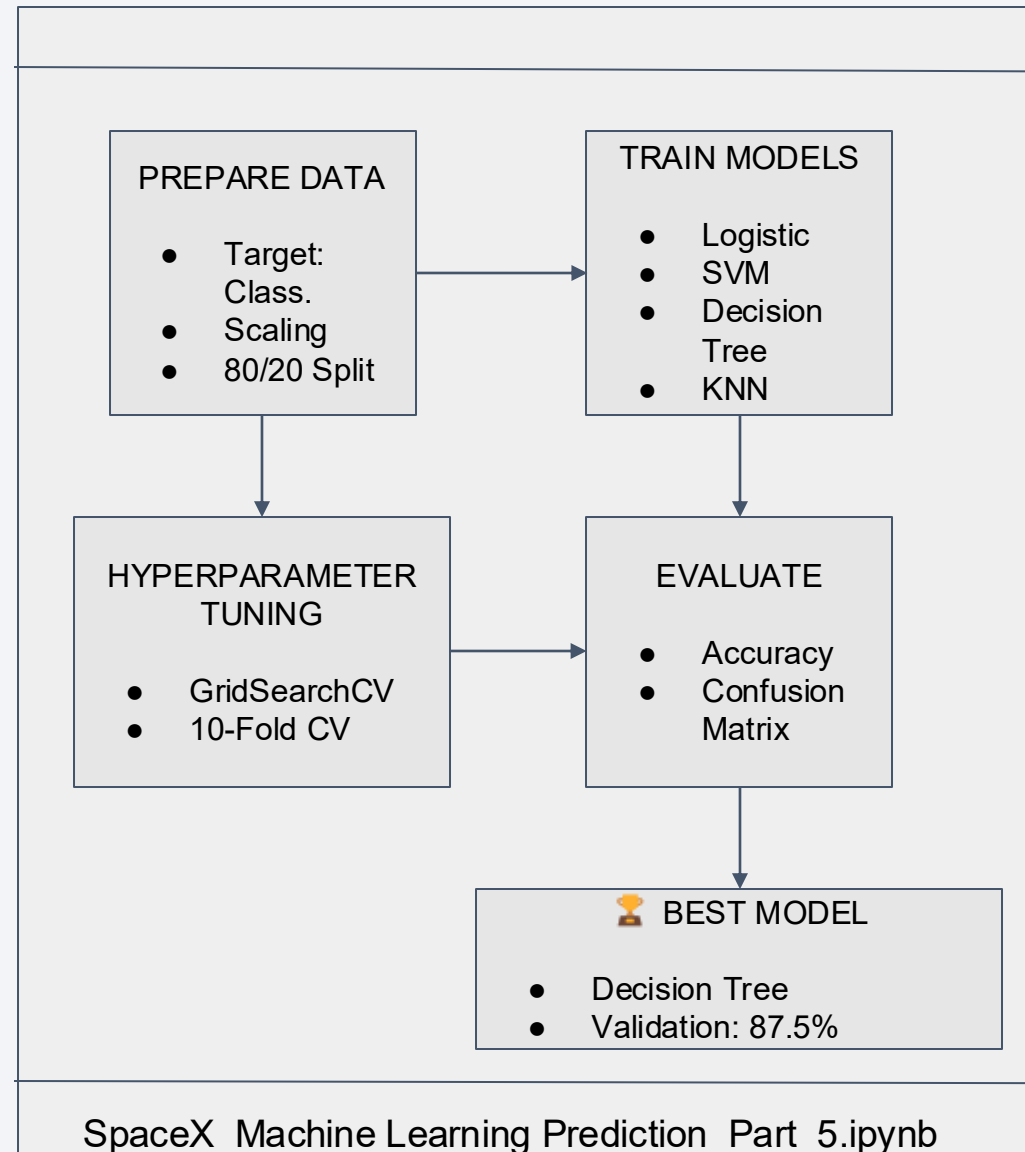


Pie Chart + Scatter Plot



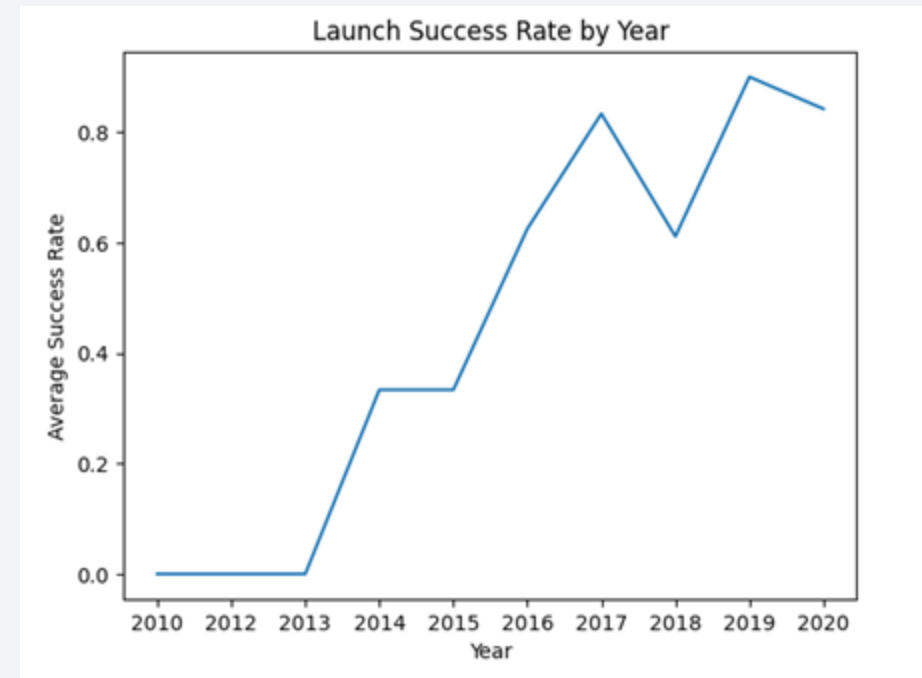
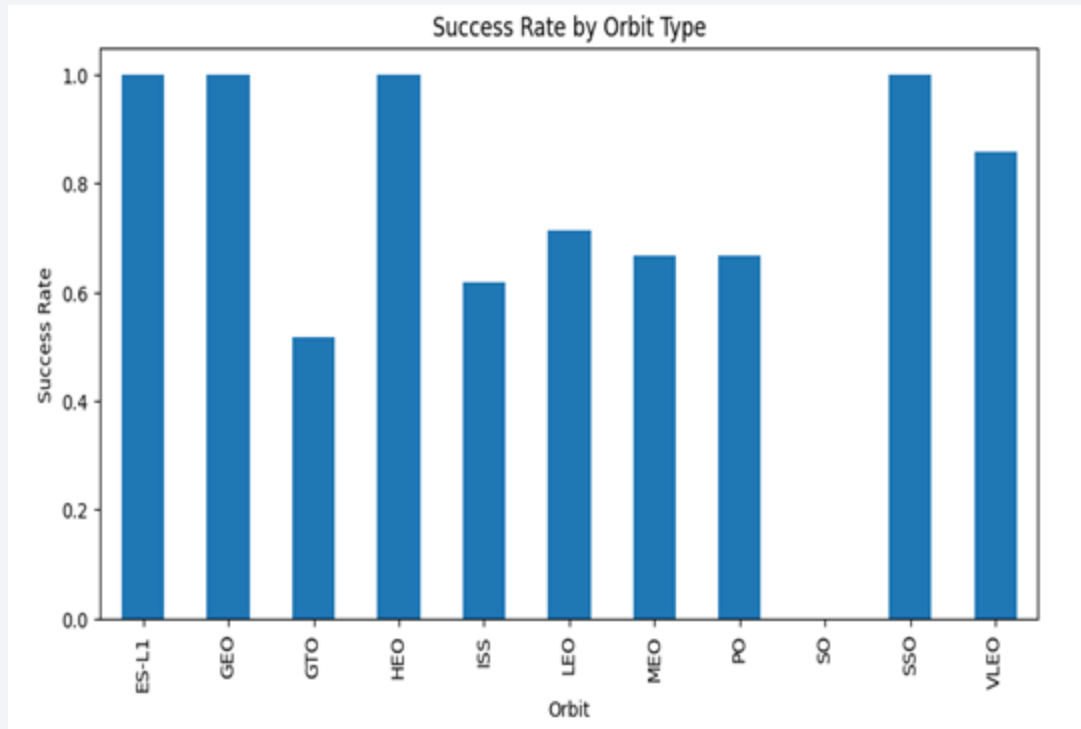
Interactive Data Analysis

Predictive Analysis (Classification)



Results

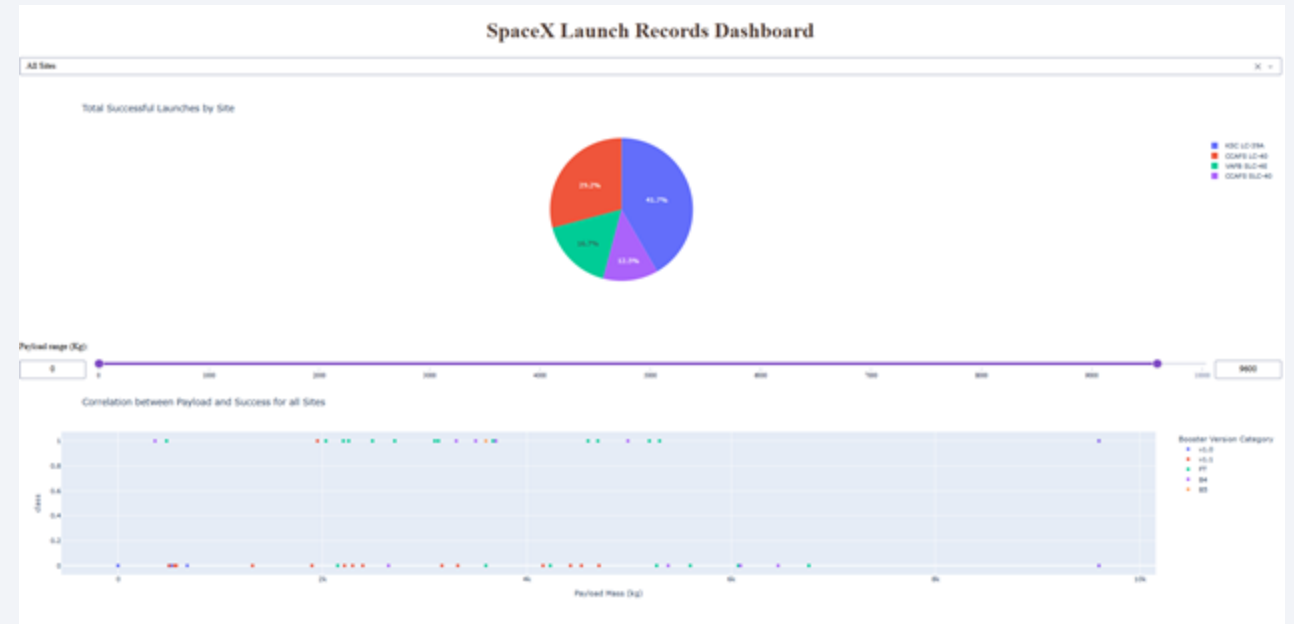
EDA Results



- Launch trends
- Site performance
- Payload patterns

Results

Interactive Analytics



- Interactive map
- Site selection
- Payload filtering

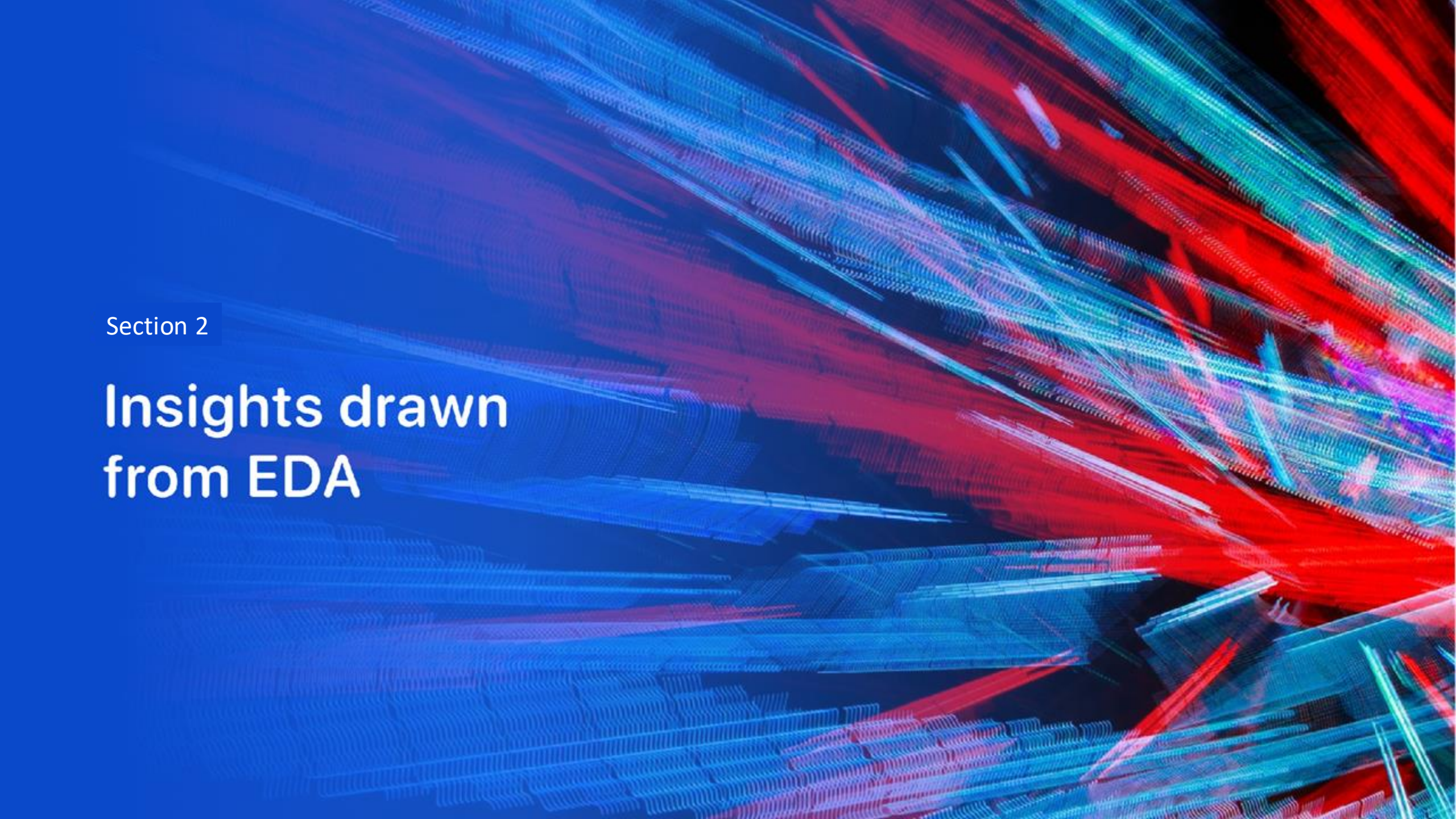
Results

Predictive Analysis

Test Performance

```
Best performing method: Logistic Regression  
Accuracy: 0.83333333333333333334
```

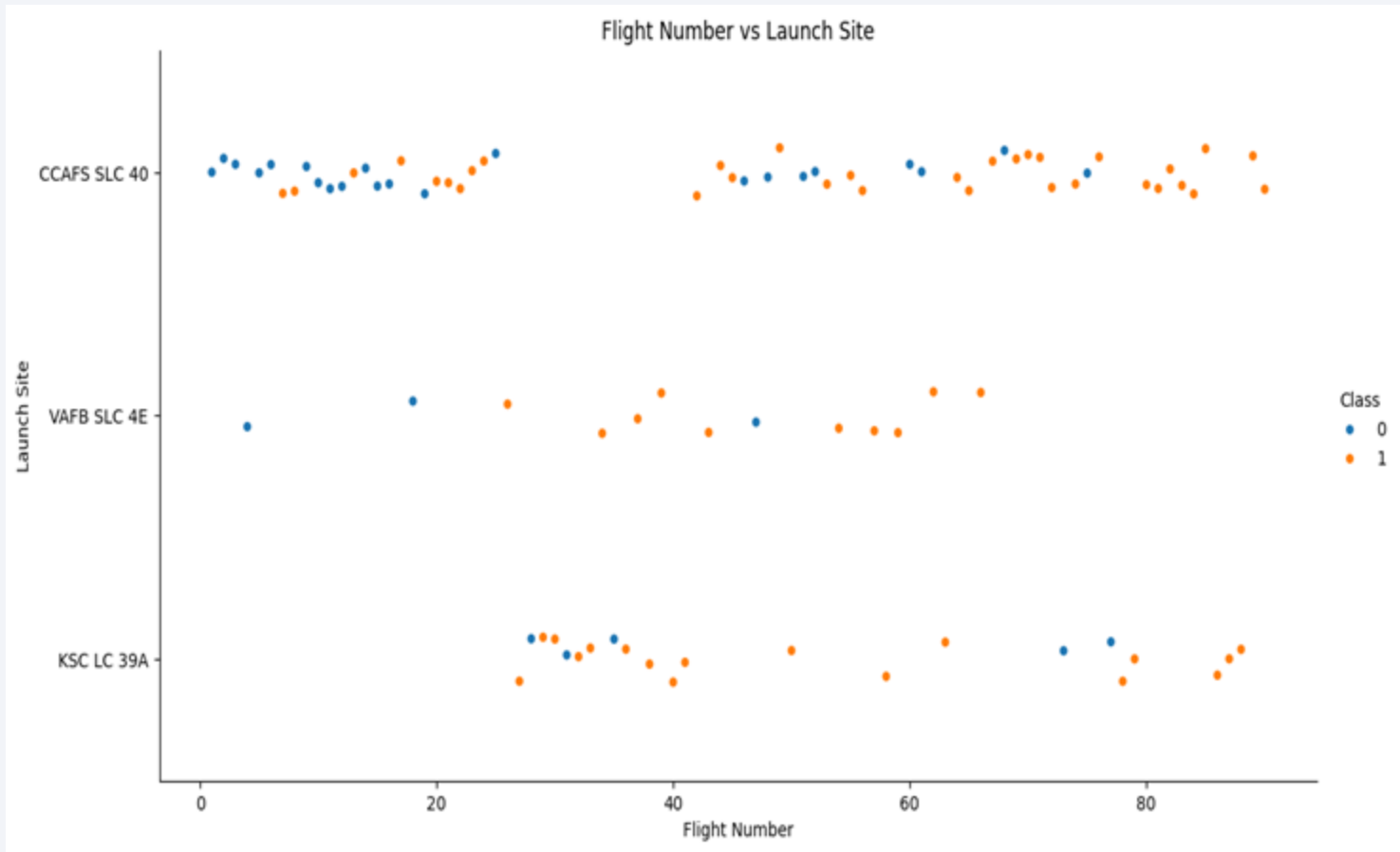
Classification models were compared using cross-validation and test performance.

The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue and red. Overlaid on these streaks is a fine, light-colored grid or mesh pattern, giving the impression of a digital or data-driven environment.

Section 2

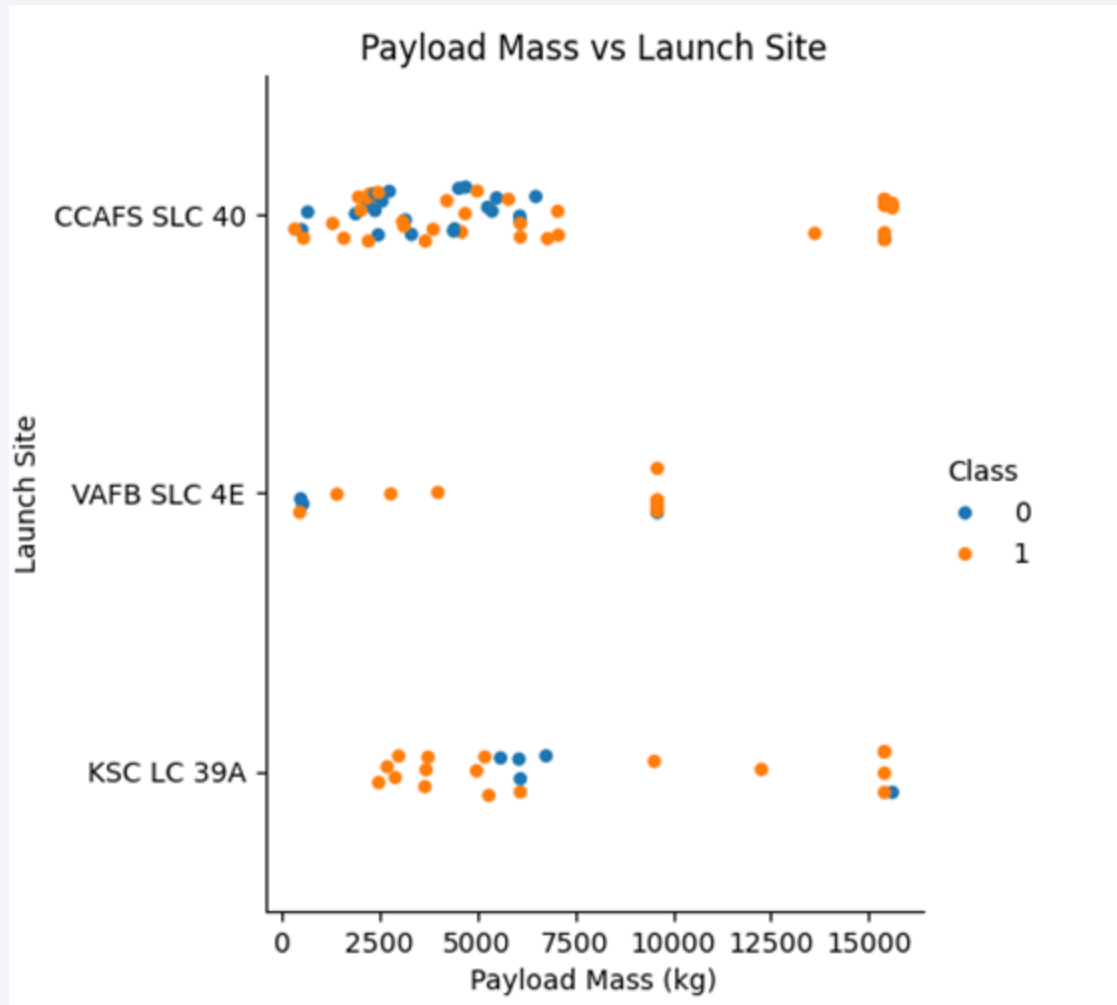
Insights drawn from EDA

Flight Number vs. Launch Site



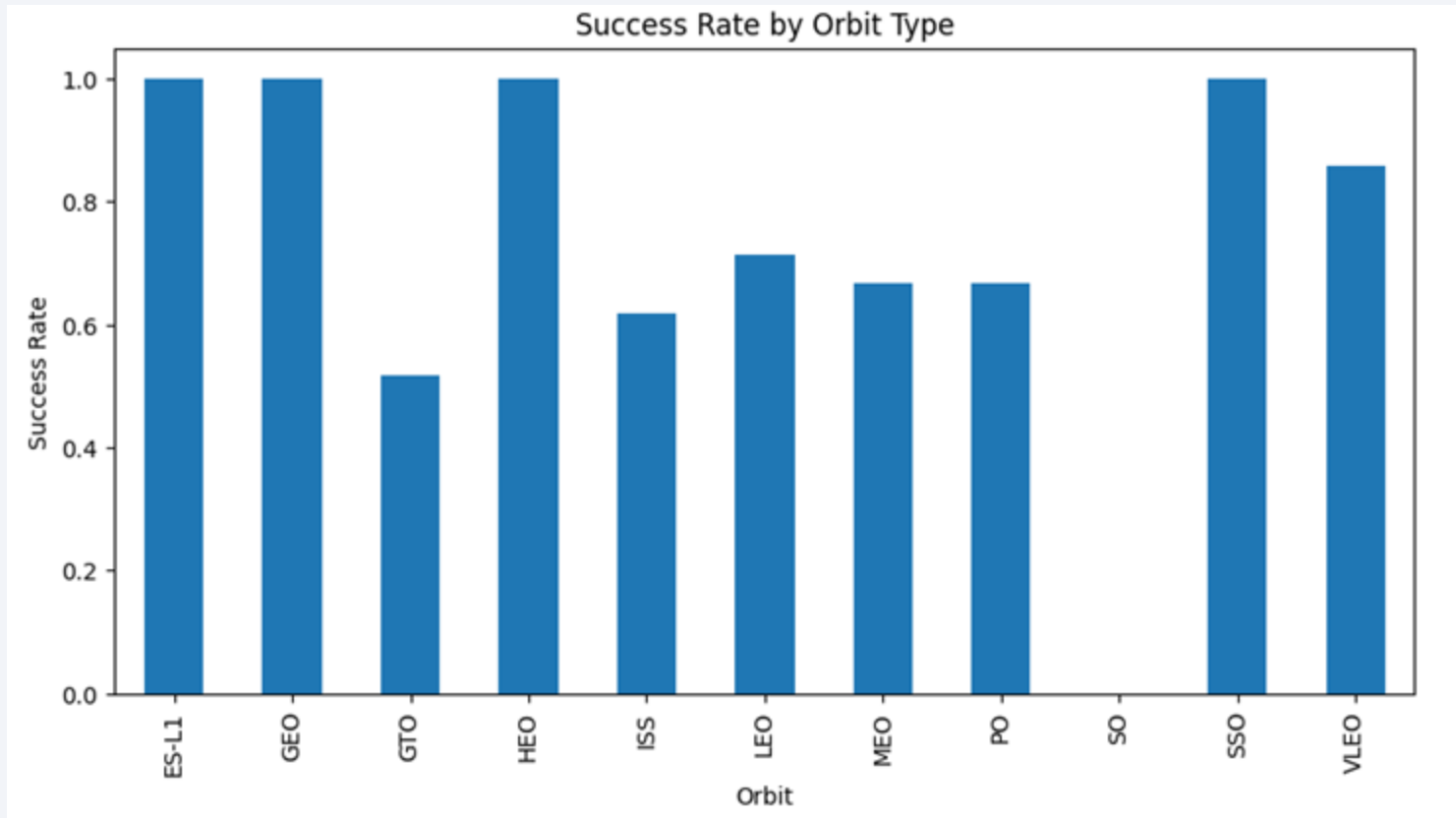
- Each point represents a SpaceX launch.
- The **Flight Number** indicates the sequence of launches.
- The **Launch Site** shows the location from which each launch took place.
- The scatter plot allows us to compare the distribution of launches across different launch sites over time.

Payload vs. Launch Site



- Each point represents a SpaceX launch and its payload mass.
- The **Payload Mass** is compared across the different **Launch Sites**.
- The plot shows that payloads vary across launch sites, with some sites handling a wider range of payload masses.
- This visualization helps identify how payload capacity is distributed among the launch sites.

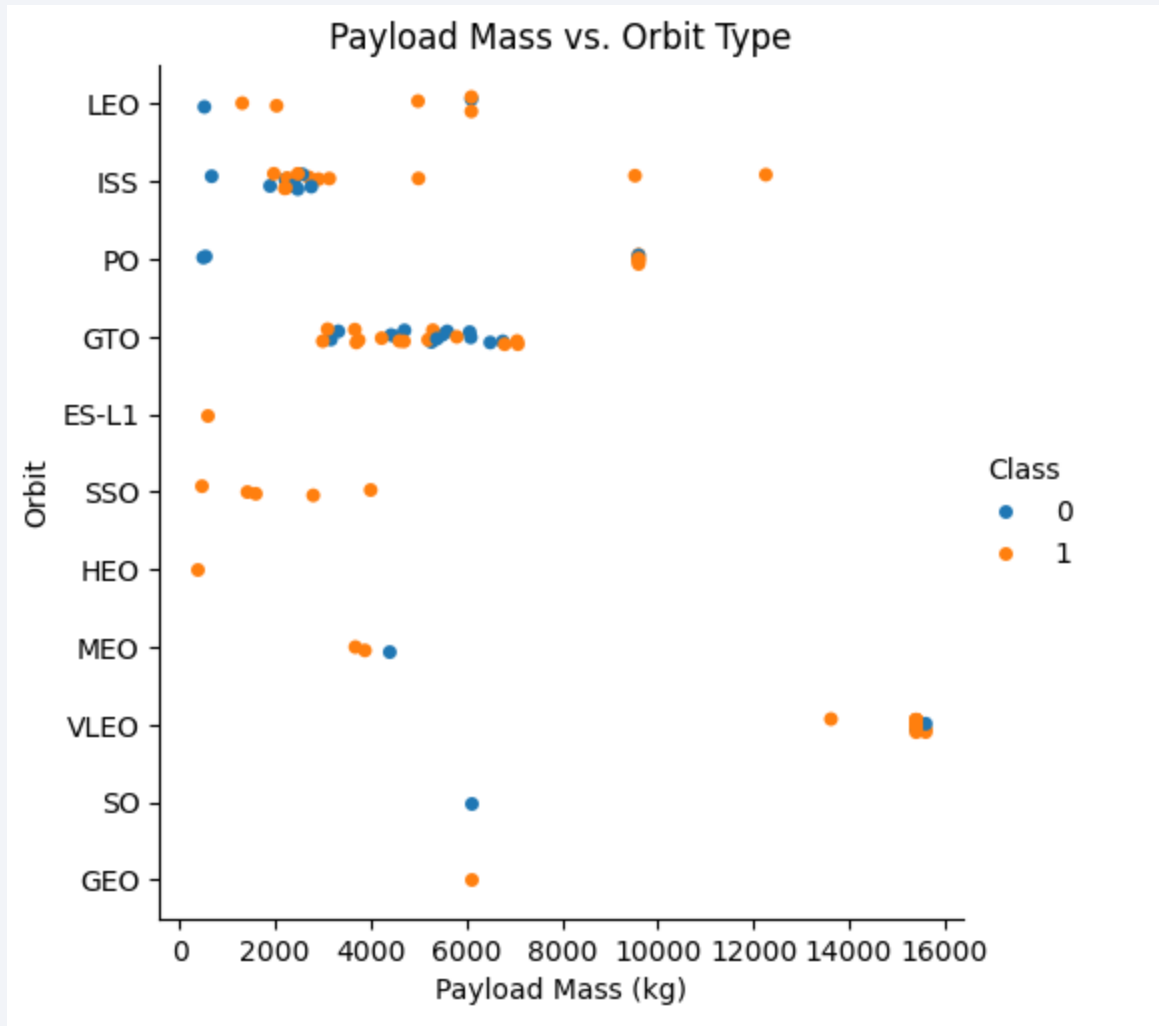
Success Rate vs. Orbit Type



- The bar chart compares the **success rate of launches across different orbit types**.
- Each bar represents an orbit type and its corresponding launch success rate.
- The visualization helps identify which orbit types have **higher or lower success rates**.
- This comparison provides insight into the relationship between the **target orbit and launch success**.

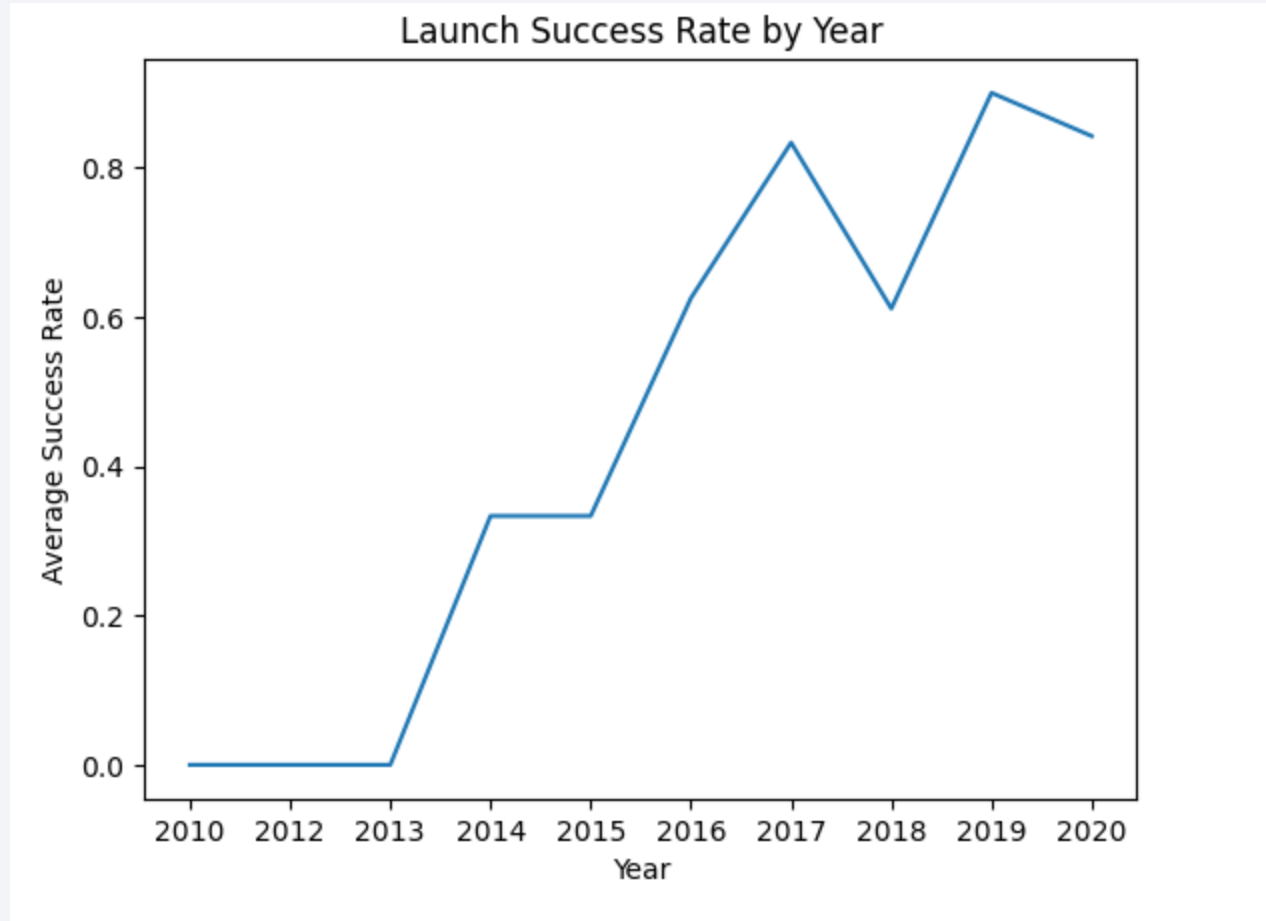
- 23

Payload vs. Orbit Type



- The scatter plot shows the relationship between **Payload Mass and Orbit Type**, with points representing successful and unsuccessful landings.
- Most launches have payload masses concentrated roughly between **2,000 and 6,000 kg**.
- **Heavier payloads** show a higher proportion of successful landings, although there are fewer launches with very high payload masses.
- The distribution of payload mass varies across orbit types, showing that **payload and orbit type are important factors when analyzing launch success**

Launch Success Yearly Trend



- The yearly success rate shows a **clear improvement over time**, especially after 2013.
- The success rate increased from approximately **33% in 2014–2015** to about **83% in 2017**.
- There were temporary decreases in **2018** and **2020**.
- The **highest success rate was reached in 2019**, at approximately **90%**.

All Launch Site Names

```
%%sql  
SELECT DISTINCT "Launch_Site"  
FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

- The query uses **DISTINCT** to identify unique launch sites.
- The analysis identifies **4 unique launch sites** in the dataset.
- These sites are located at CCAFS, KSC, and VAFB.
- This result provides an overview of the launch locations used in the SpaceX missions analyzed.

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- The **LIKE** 'CCA%' condition filters launch sites that begin with “CCA”.
- **LIMIT 5** restricts the query to the first **5 matching records**.
- All five records correspond to the **CCAFS LC-40** launch site.
- The results represent early Falcon 9 missions from **2010 to 2013**, mainly targeting LEO and LEO (ISS).

Total Payload Mass

Total Payload Mass (kg)

45596

- The **WHERE** clause filters the dataset to **NASA (CRS)** missions.
- The **SUM()** function calculates the total payload mass carried by these missions.
- The total payload mass is **45,596 kg**.
- This result represents the combined payload mass delivered through NASA (CRS) missions in the analyzed dataset.

Average Payload Mass by F9 v1.1

Average Payload Mass (kg)

2928.4

- The **WHERE** clause filters the data to the **F9 v1.1** booster version.
- The **AVG()** function calculates the average payload mass.
- The average payload mass carried by F9 v1.1 was approximately **2,928.4 kg**.
- This result provides a reference for the typical payload capacity of this booster version.

First Successful Ground Landing Date

First Successful Ground Pad Landing

2015-12-22

- The **WHERE** clause filters the records to **successful ground pad landings**.
- The **MIN()** function identifies the earliest landing date.
- The first successful ground pad landing occurred on **December 22, 2015**.
- This milestone marked an important step toward reusable Falcon 9 boosters.

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

- The query filters for **successful drone ship landings**.
- The payload mass is restricted to values **greater than 4,000 kg and less than 6,000 kg**.
- Four booster versions meet both conditions: **F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2**.
- These results demonstrate successful recovery of boosters carrying relatively heavy payloads.

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- The query counts mission outcomes classified as **Success** or **Failure**.
- The results show **100 successful missions** and **1 failed mission**.
- Successful missions represent the vast majority of the analyzed launches.
- This indicates a very high overall mission success rate in the dataset.

Boosters Carried Maximum Payload

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

- A subquery is used to identify the **maximum payload mass** in the dataset.
- The main query returns the booster versions associated with that maximum payload.
- The **DISTINCT** keyword ensures that only unique booster versions are displayed.
- The results identify **12 booster versions** that carried the maximum payload mass.

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- The query filters launches from **2015** with a “**Failure (drone ship)**” landing outcome.
- Two failed drone ship landing attempts were identified.
- The failed boosters were **F9 v1.1 B1012** and **F9 v1.1 B1015**.
- Both launches were performed from **CCAFS LC-40**, in January and April 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing_Outcome	Count	Rank
No attempt	10	1
Success (drone ship)	5	2
Failure (drone ship)	5	2
Success (ground pad)	3	4
Controlled (ocean)	3	4
Uncontrolled (ocean)	2	6
Failure (parachute)	2	6
Precluded (drone ship)	1	8

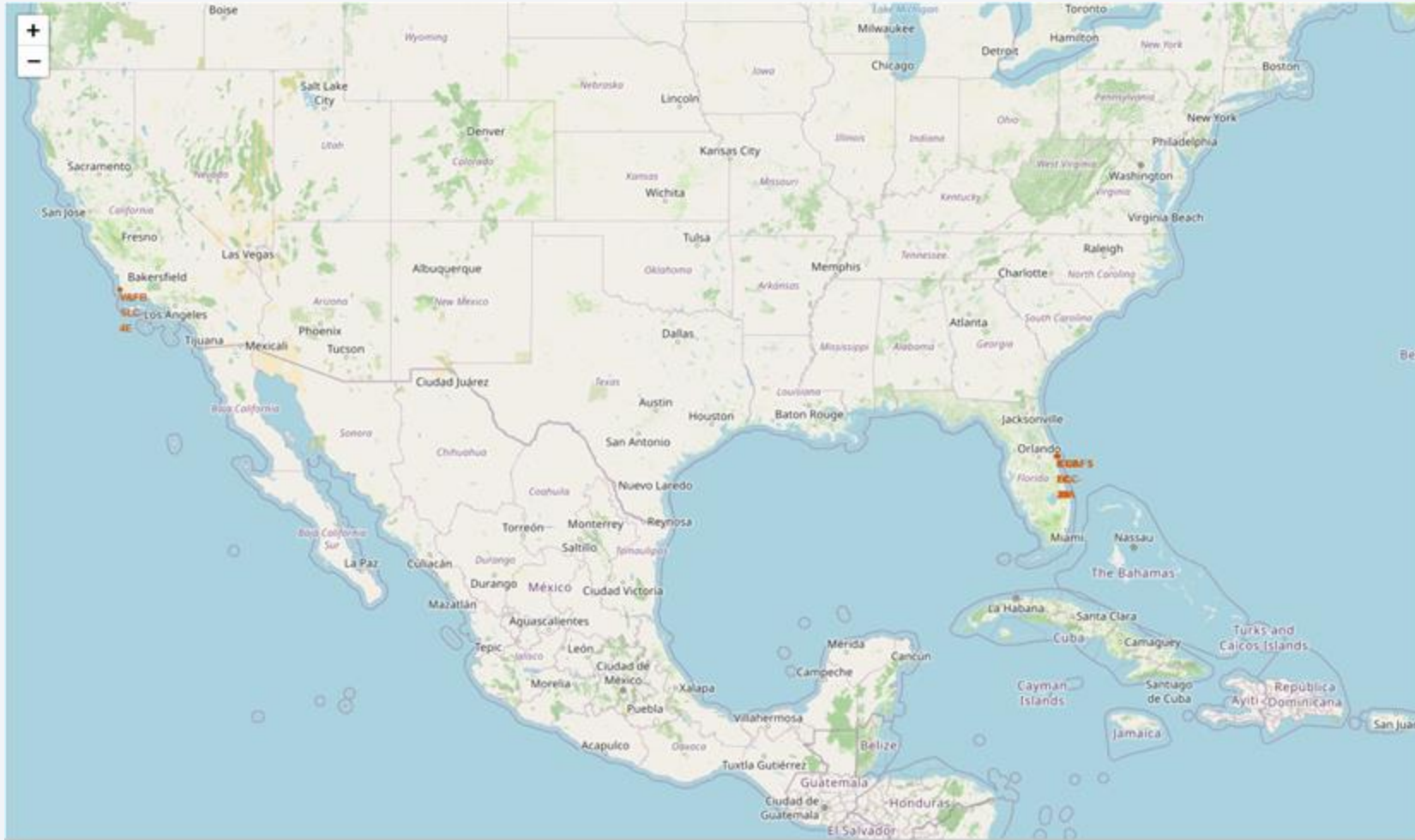
- The query filters landing outcomes between **June 4, 2010 and March 20, 2017**.
- The outcomes are grouped and ranked in **descending order by count**.
- **“No attempt”** is the most frequent outcome, with **10 records**.
- **Success and Failure (drone ship)** each have **5 records**, followed by other landing outcomes with lower frequencies.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear blue sky.

Section 3

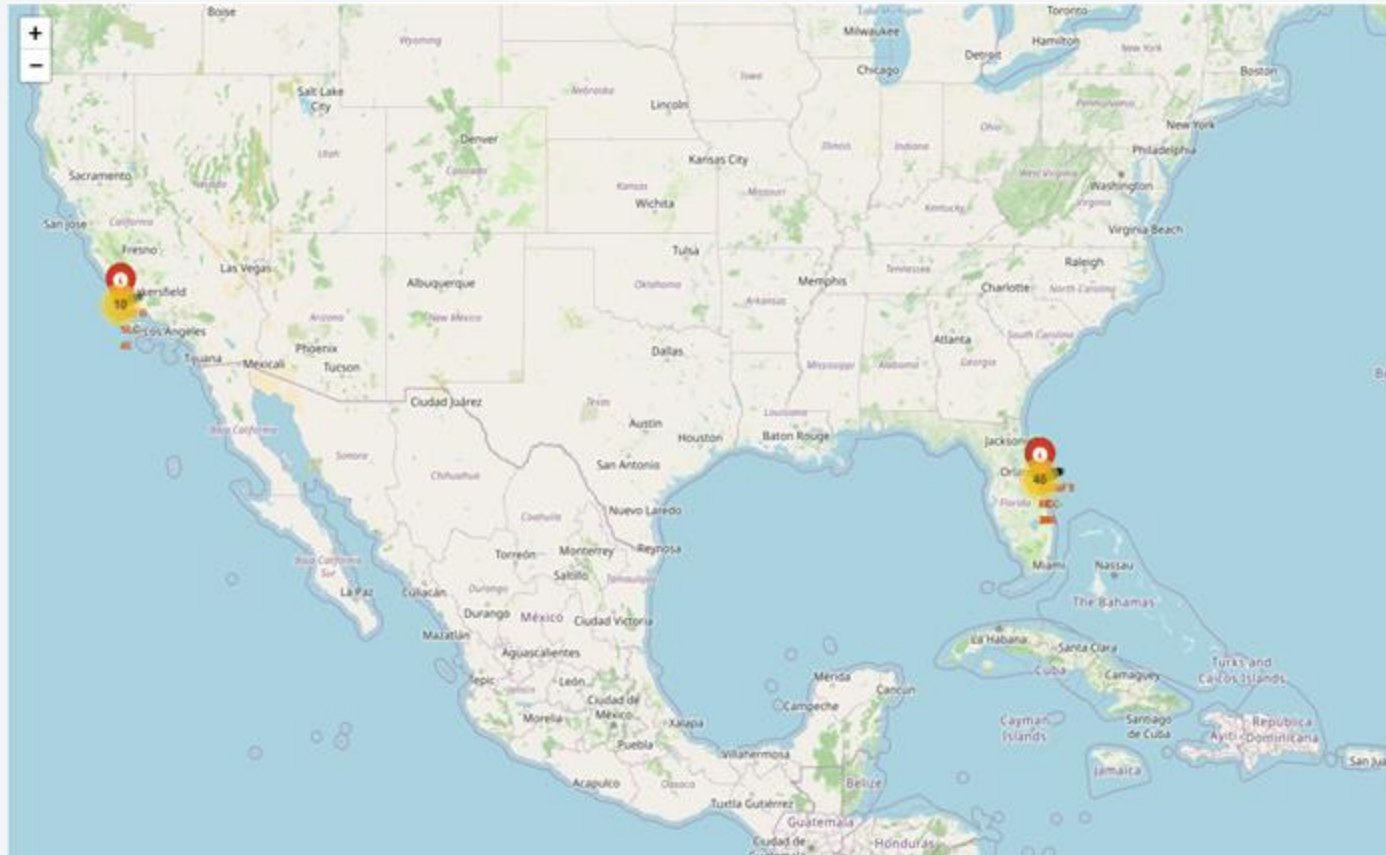
Launch Sites Proximities Analysis

SpaceX Launch Sites – Global Location Map



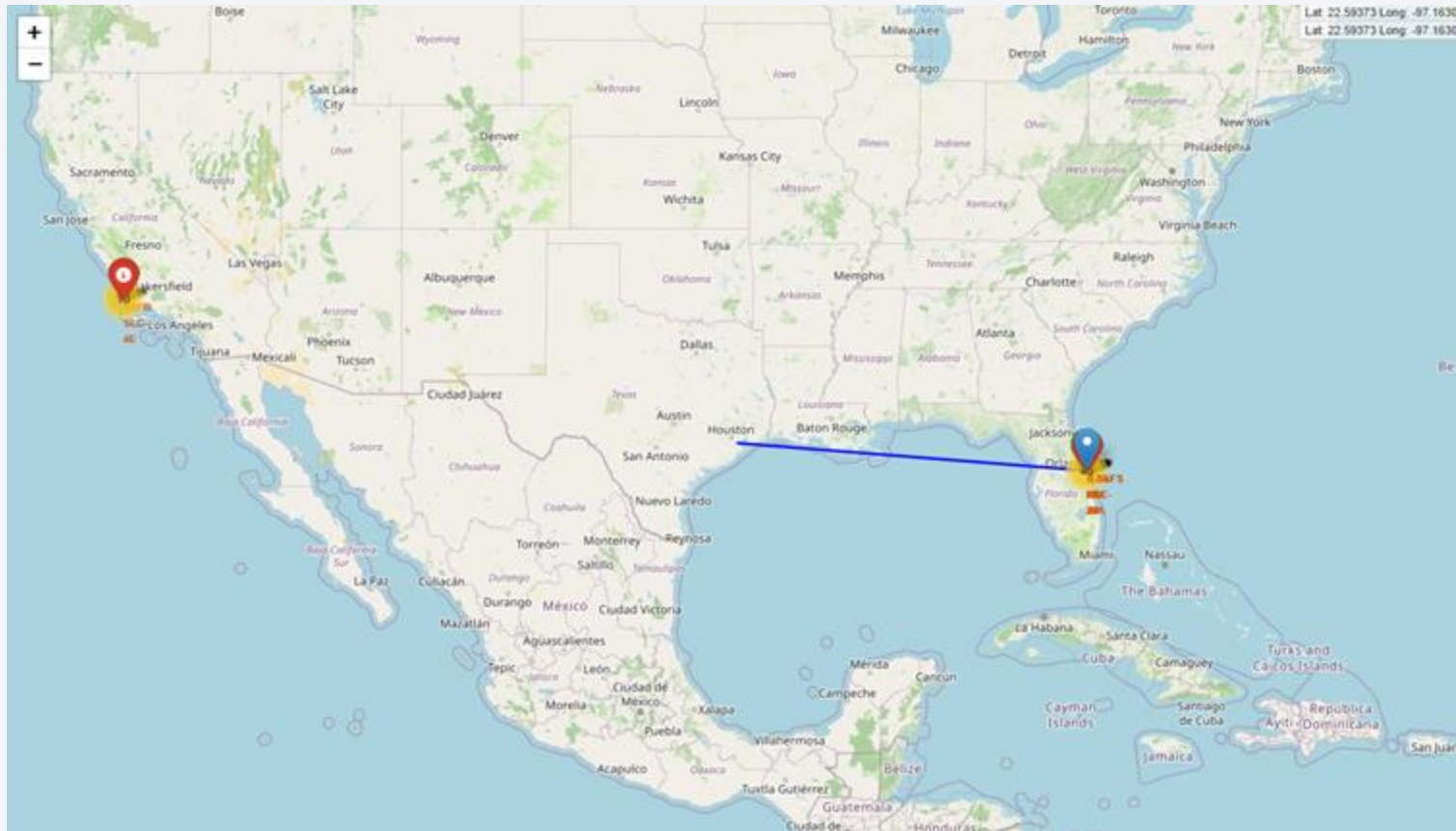
- The Folium map displays the geographic location of all four SpaceX launch sites analyzed in the dataset.
- The launch sites are concentrated along the U.S. coastline, specifically in **Florida and California**.
- Three launch sites are located in Florida: **CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A**.
- **VAFB SLC-4E** is located on the California coast.
- The map provides a clear geographic view of the launch-site distribution and supports further analysis of site proximity and launch success.

SpaceX Launch Outcomes by Launch Site

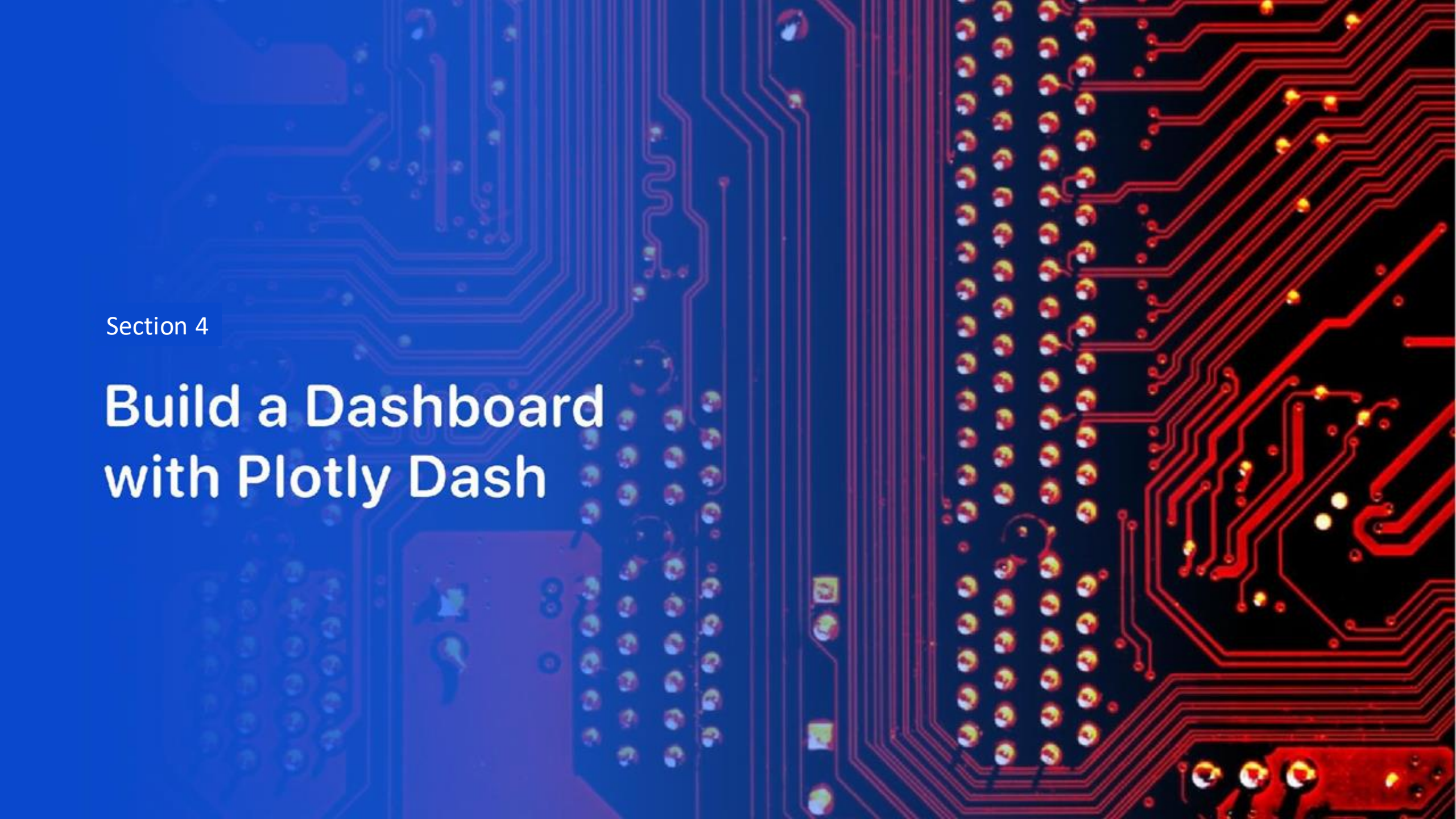


- The map displays individual SpaceX launch outcomes for each launch site using color-coded markers.
- **Green markers represent successful launches**, while **red markers represent failed launches**.
- A **MarkerCluster** is used to group launches with the same geographic coordinates and improve map readability.
- The distribution of green and red markers allows comparison of launch outcomes across the different sites.
- The map provides an interactive geographic view of launch success and failure patterns.

Launch Site Proximity Analysis – Folium



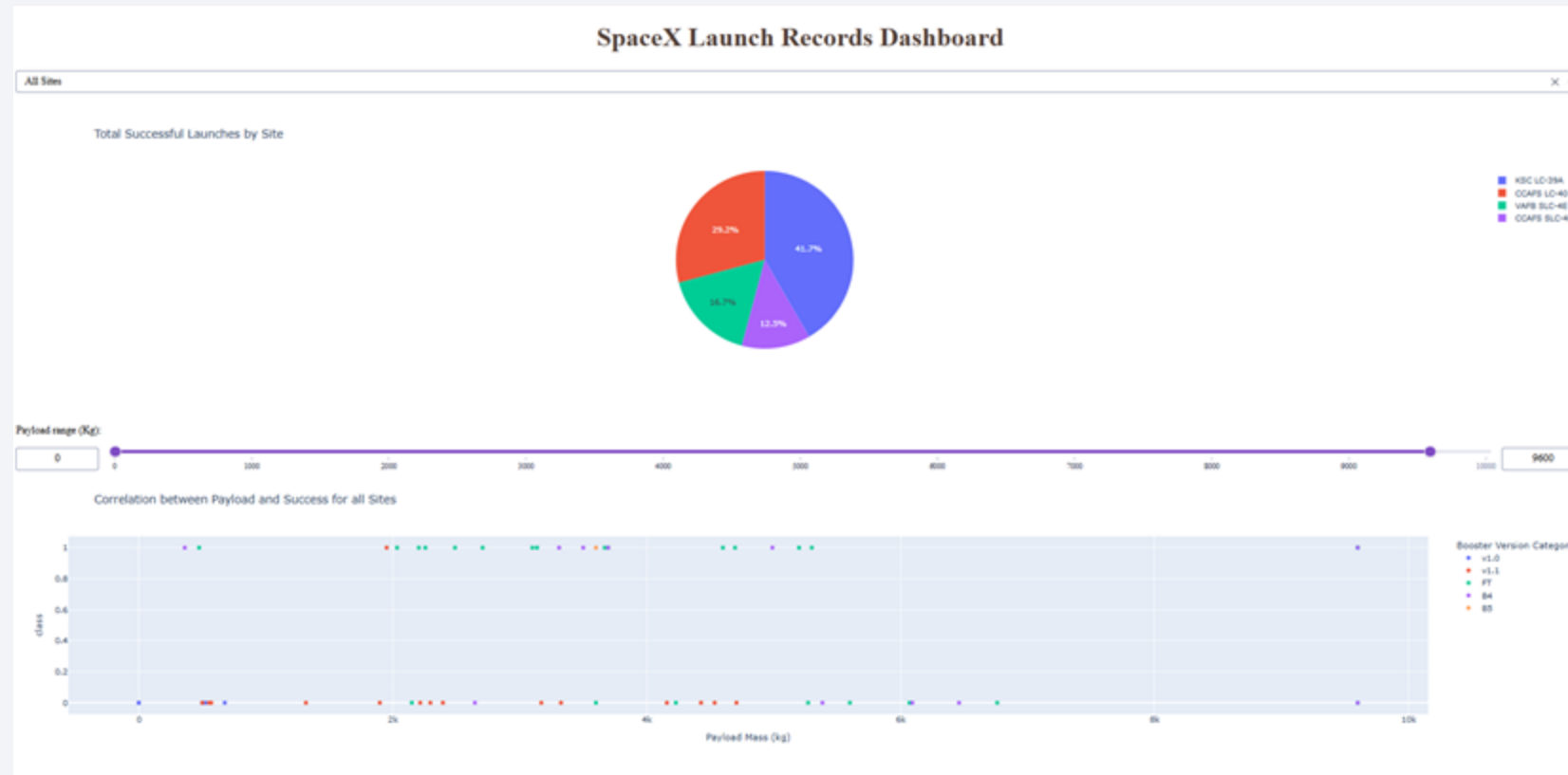
- The map analyzes the proximity of a selected SpaceX launch site to nearby geographic and transportation features.
- Distances are calculated from the launch site to the **coastline, railway, highway, and nearest city**.
- The orange distance labels display the calculated distance in **kilometers**, while the lines show the relationship between the launch site and each point.
- The analysis shows that launch sites are located close to transportation infrastructure and coastlines, while maintaining greater distance from populated areas.
- This geographic configuration supports transportation access and helps reduce the potential risk to densely populated areas.



Section 4

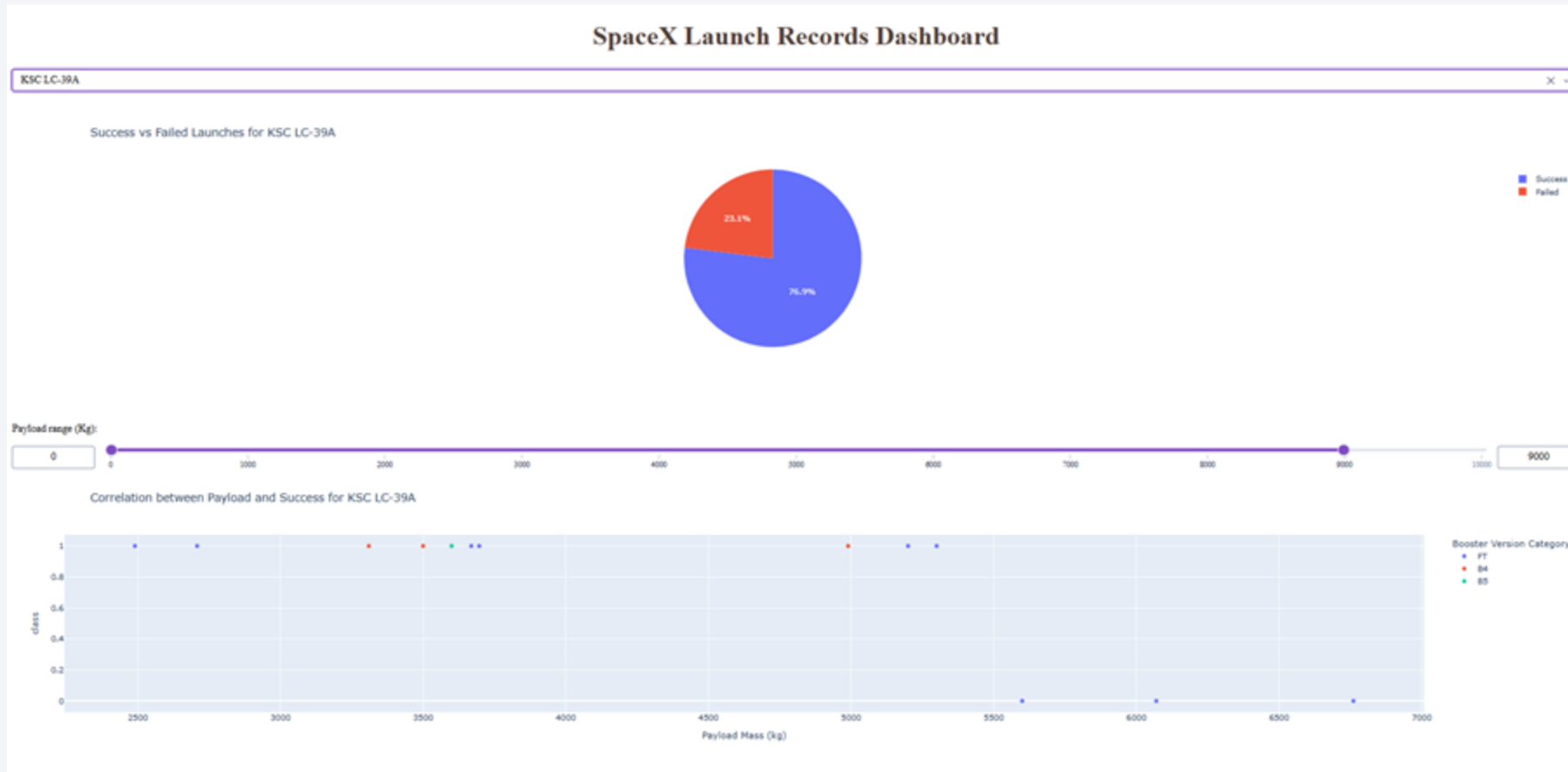
Build a Dashboard with Plotly Dash

Launch Success Count by Site



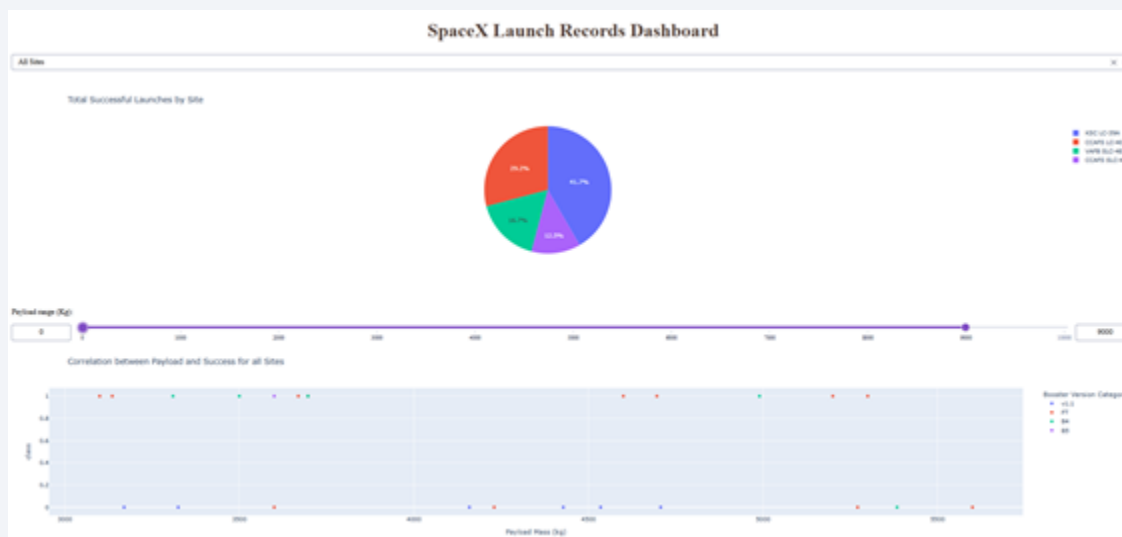
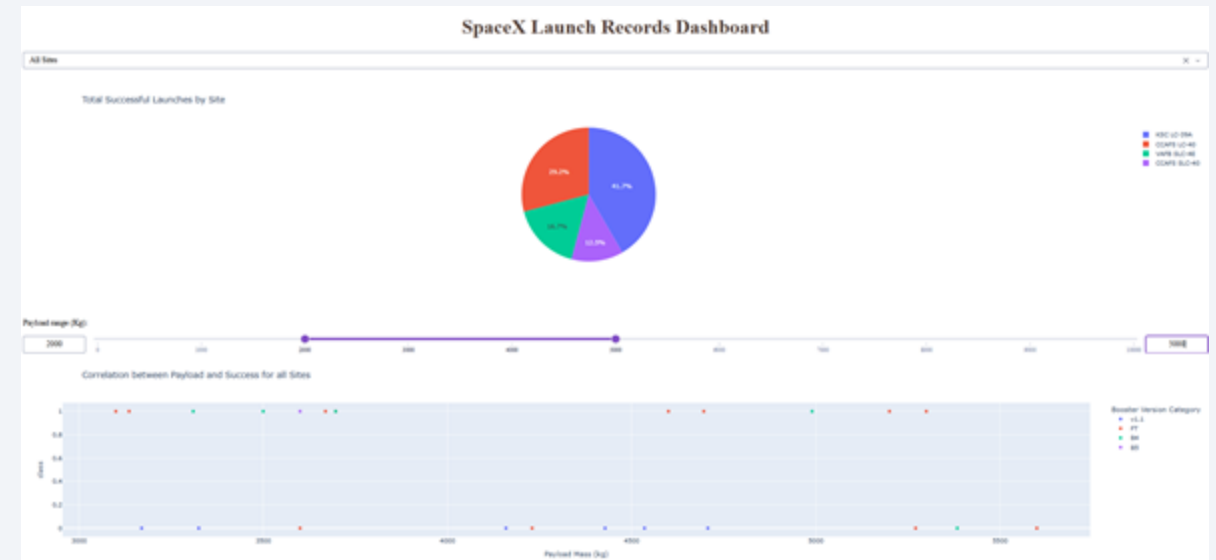
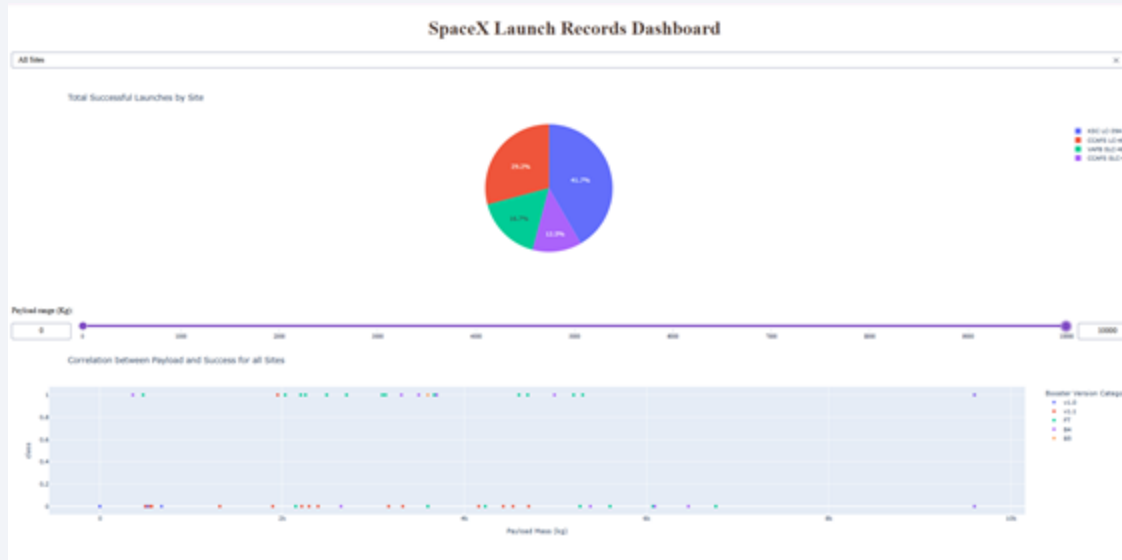
- The pie chart shows the distribution of successful Falcon 9 launches across all launch sites.
- KSC LC-39A** has the highest share of successful launches, accounting for **41.7%** of the total.
- CCAFS LC-40** follows with **29.2%**, while **VAFB SLC-4E** and **CCAFS SLC-40** account for **16.7%** and **12.5%**, respectively.
- KSC LC-39A therefore contributed the largest number of successful launches among the analyzed sites.

Launch Success Ratio – KSC LC-39A



- **KSC LC-39A** achieved the highest launch success ratio among all launch sites.
- **76.9%** of launches were successful, while **23.1%** were unsuccessful.
- The pie chart clearly shows that successful landings represent the majority of outcomes at this launch site.
- KSC LC-39A had **10 successful landings and 3 failed landings**, indicating the strongest landing performance among the analyzed sites.

Payload Mass vs. Launch Outcome Analysis



- The scatter plot shows the relationship between **payload mass and launch outcome** across all launch sites.
- The **1,952–5,300 kg** payload range shows the **highest launch success rate** in the analyzed data.
- Payloads **below 1,952 kg and above 5,300 kg** show comparatively lower success rates.
- The **FT and B5 booster versions** show the highest success performance among the analyzed booster categories.
- The Range Slider enables interactive filtering of payload mass, making it easier to identify success patterns across different payload ranges.



Section 5

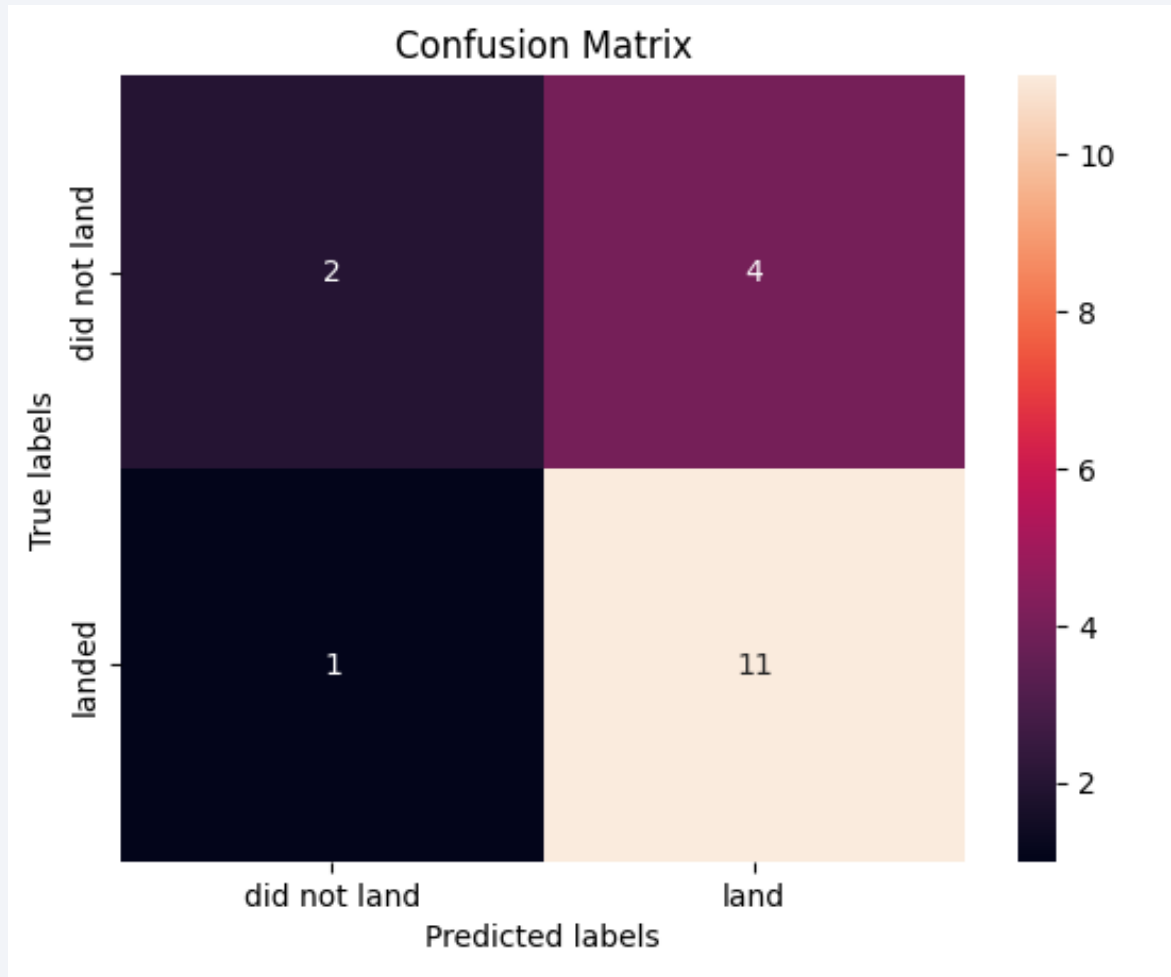
Predictive Analysis (Classification)

Classification Accuracy



- Four classification models were evaluated: Logistic Regression, SVM, Decision Tree, and KNN.
- The **Decision Tree Classifier** achieved the highest accuracy at **87.50%**.
- SVM and KNN achieved **84.82%**, while Logistic Regression achieved **84.64%**.
- Therefore, the **Decision Tree** was selected as the best-performing classification model.

Confusion Matrix– Decision Tree Classifier



- The Decision Tree classifier was the best-performing model based on the validation accuracy.
- The confusion matrix compares the **actual landing outcome** with the **predicted outcome**.
- Correct predictions appear on the diagonal, while incorrect predictions represent false positives and false negatives.
- The matrix shows how effectively the model distinguishes between successful and unsuccessful Falcon 9 landings.

Conclusions

- **Point 1:**  Multiple factors influence launch success.
- **Point 2:**  Data preprocessing improved model reliability.
- **Point 3:**  Launch sites showed different success patterns.
- **Point 4:**  Machine learning enabled launch outcome prediction.
- **Point 5:**  Visualizations revealed important data patterns.
- **Point 6:**  Data science supports better launch decisions.

Appendix 1 — Data Preparation

```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'
```

We should see that the request was successful with the 200 status response code

```
response=requests.get(static_json_url)
```

```
response.status_code
```

200

Now we decode the response content as a json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
# Use json_normalize method to convert the json result into a dataframe
data_json = response.json()
data = pd.json_normalize(data_json)
```

Task 3: Dealing with Missing Values

Calculate below the mean for the `PayloadMass` using the `.mean()`. Then use the mean and the `.replace()` function to replace `np.nan` values in the data with the mean you calculated.

```
# Calculate the mean value of PayloadMass column
mean_payload_mass = data_falcon9['PayloadMass'].mean()

print(mean_payload_mass)

# Replace the np.nan values with its mean value
data_falcon9['PayloadMass'] = data_falcon9['PayloadMass'].replace(
    np.nan,
    mean_payload_mass
)

data_falcon9.isnull().sum()
```

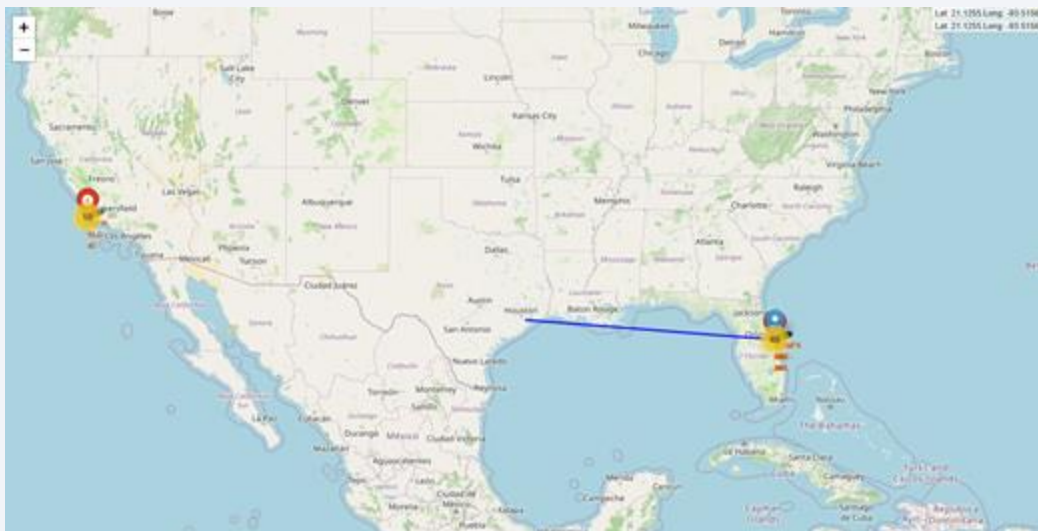
1204.959411764707

FlightNumber	0
Date	0
BoosterVersion	0
PayloadClass	0
Orbit	0
LaunchSite	0
Outcome	0
Flights	0
GridFins	0
Reused	0
Legs	0
LandingPad	26
Block	0
ReusedCount	0
Serial	0
Longitude	0
Latitude	0
dtype: int64	

[illegible]

- Screenshot of the data extraction code.
- Screenshot of head() / the dataset.
- Screenshot of missing value handling.

Appendix 2 — Geospatial Analysis



- Launch sites map.
- Success/failure map.
- Distance-to-coast map..

Appendix 3 — Machine Learning

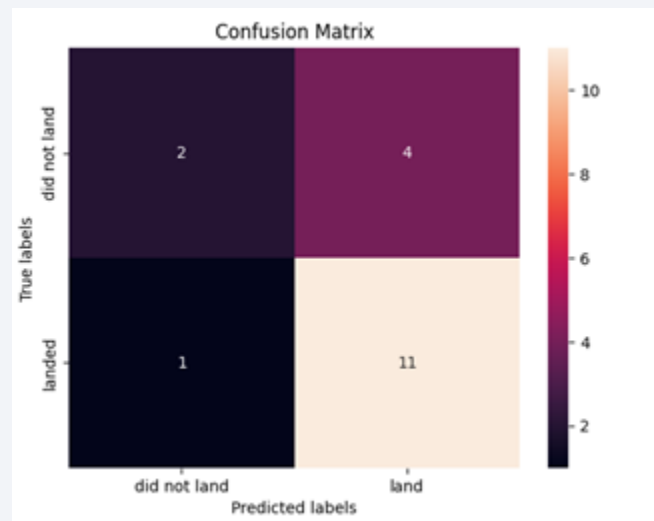
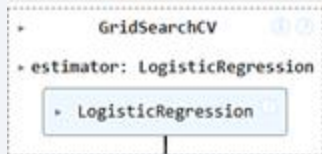
Create a logistic regression object then create a GridSearchCV object `logreg_cv` with `cv = 10`. Fit the object to find the best parameters from the dictionary `parameters`.

```
parameters = {'C':[0.01,0.1,1],
              'penalty':['l2'],
              'solver':['lbfgs']}

parameters = {'C':[0.01,0.1,1], 'penalty':['l2'], 'solver':['lbfgs']}# l1 lasso l2 ridge
lr=LogisticRegression()

logreg_cv = GridSearchCV(
    lr,
    parameters,
    cv=10
)

logreg_cv.fit(X_train, Y_train)
```



Find the method performs best:

```
print("Logistic Regression:", logreg_cv.score(X_test, Y_test))
print("SVM:", svm_cv.score(X_test, Y_test))
print("Decision Tree:", tree_cv.score(X_test, Y_test))
print("KNN:", knn_cv.score(X_test, Y_test))
```

```
Logistic Regression: 0.8333333333333334
SVM: 0.8333333333333334
Decision Tree: 0.7222222222222222
KNN: 0.8333333333333334
```

```
models = {
    "Logistic Regression": logreg_cv.score(X_test, Y_test),
    "SVM": svm_cv.score(X_test, Y_test),
    "Decision Tree": tree_cv.score(X_test, Y_test),
    "KNN": knn_cv.score(X_test, Y_test)
}
```

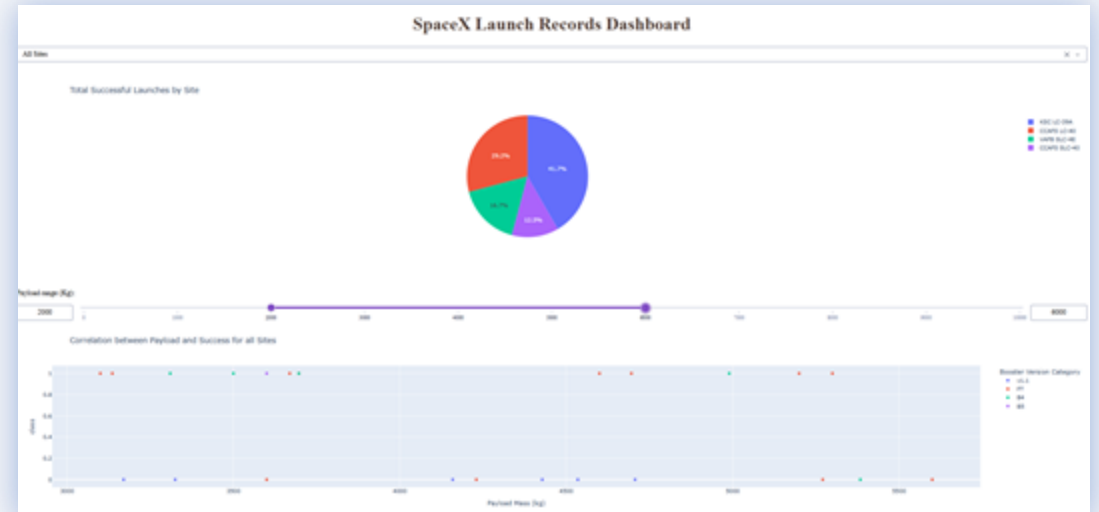
```
best_model = max(models, key=models.get)
```

```
print("Best performing method:", best_model)
print("Accuracy:", models[best_model])
```

```
Best performing method: Logistic Regression
Accuracy: 0.8333333333333334
```

- Logistic Regression / Random Forest code.
- Accuracy comparison.
- Matrix Confusion.

Appendix 4 — Interactive Visualization



- Scatter plot of payload mass vs. launch outcome.
- Screenshot of the slider in different positions.

Inclusion of innovative ideas

- Multiple Data Sources
- Geospatial Analysis
- Interactive Dashboard
- Predictive Machine Learning
- Reproducible Data Science



“The project goes beyond descriptive analysis by combining interactive visualization, geospatial analysis, and predictive machine learning to generate actionable insights.”

Thank you!

